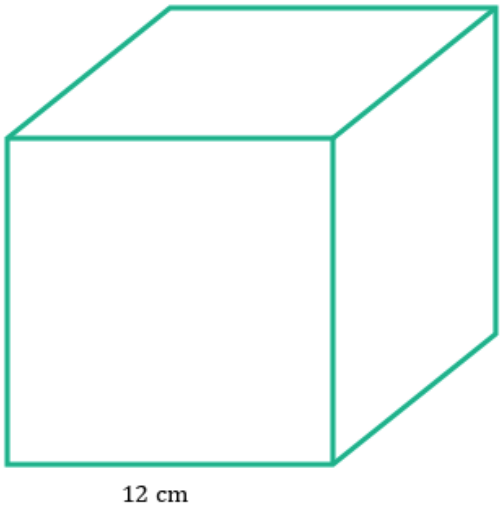
Volume of prisms and cylinders

1 Find the volume of the cube shown.



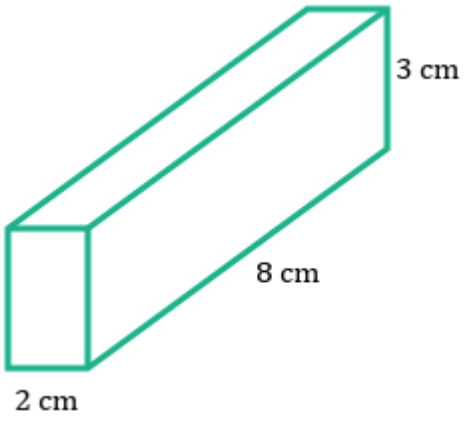
2 Find the side length of a cube with volume $27\ \text{cm}^3$.

3 Find the length of the rectangular prism with volume $162\ \text{cm}^3$, width $6\ \text{cm}$ and height $3\ \text{cm}$.

4 A pentagonal prism has a volume of $176.4\ \text{m}^3$ and a cross sectional area of $29.4\ \text{m}^2$.
Find the height of the pentagonal prism.

5 Find the volume of the following prisms:

a

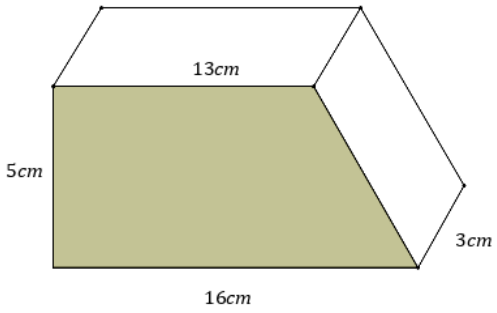


b

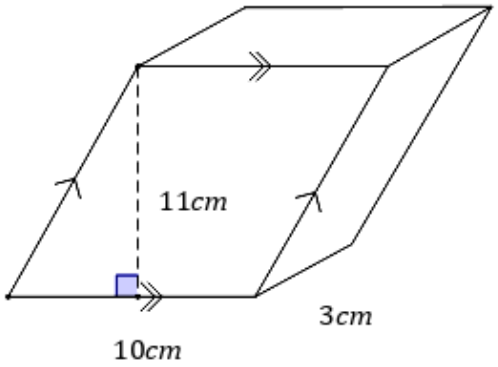


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

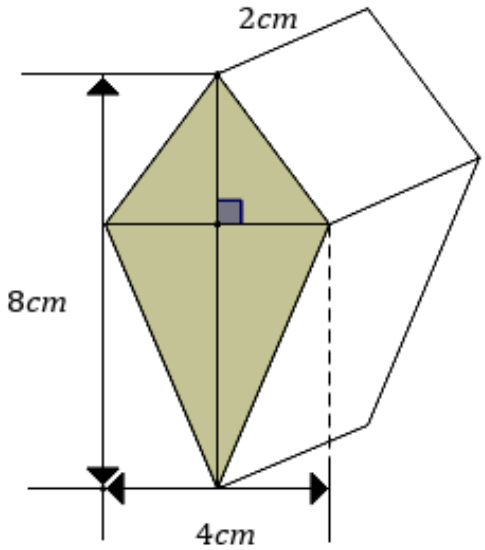
c



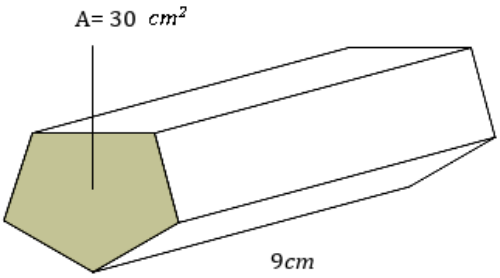
d



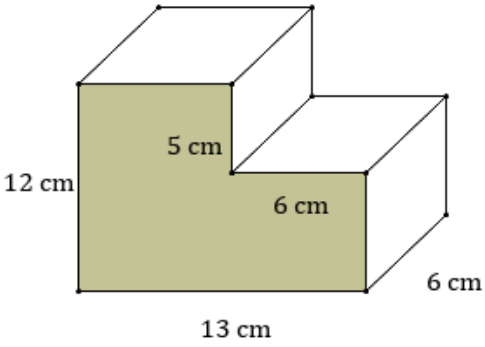
e



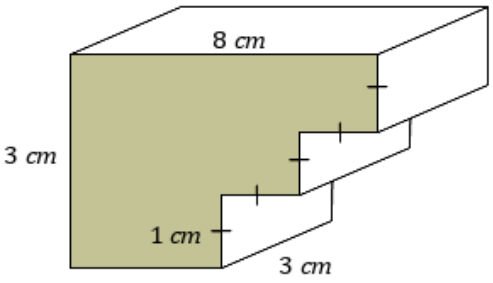
f



g

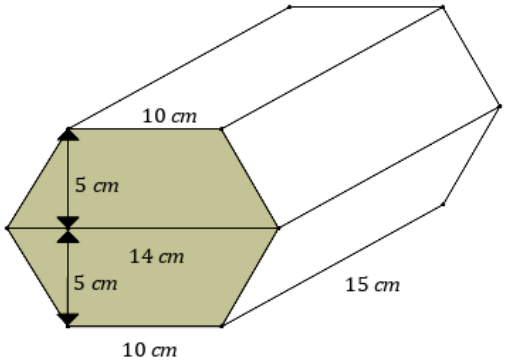


h

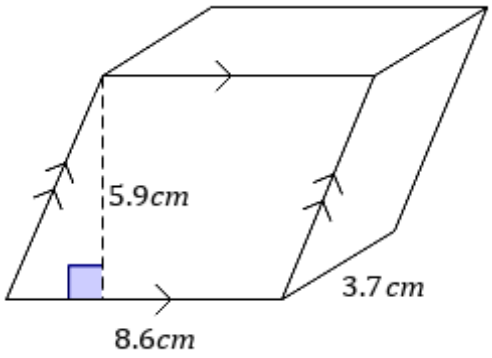


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

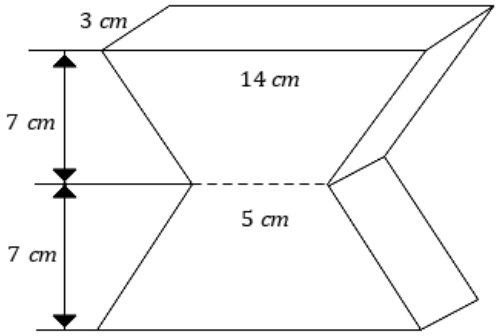
i



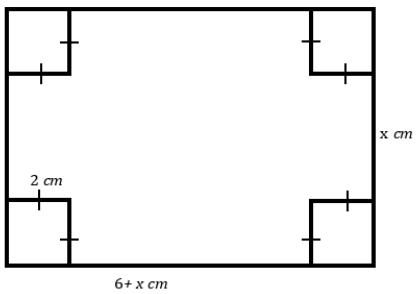
j



k



6 The length of a rectangular piece of cardboard is 6 cm longer than its width. Squares with sides of length 2 cm are cut from the four corners, and the flaps are folded upward to form an open box with a volume of 32 cm^3.
Determine whether following equations correctly describes the volume of the box.



a $x\,(x+6)=32$

b $x\,(x+6)\,(2)=32$

c $(x+2)\,(x-4)=32$

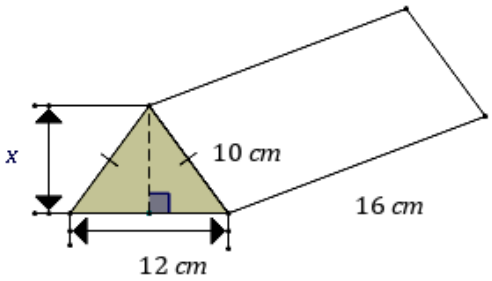
d $(x+2)\,(x-4)\,(2)=32$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Consider the following diagram.

a Find the value of the unknown height x.

b Hence find the volume of the solid.

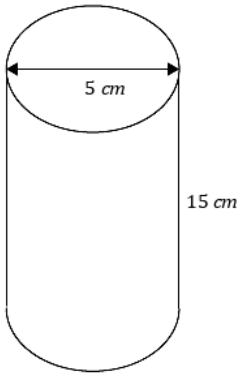


8 Find the volume of a cylinder correct to one decimal place if its diameter is 2 cm and its height is 19 cm.

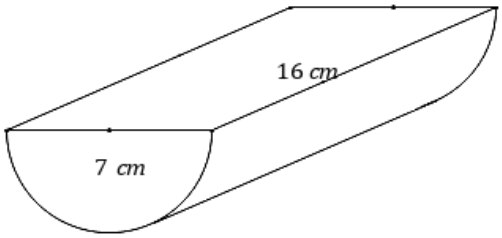
9 Find the volume of a cylinder with radius 7 cm and height 15 cm, correct to two decimal places.

10 Find the volume of the following solids, correct to one decimal place. All measurements are in centimeters.

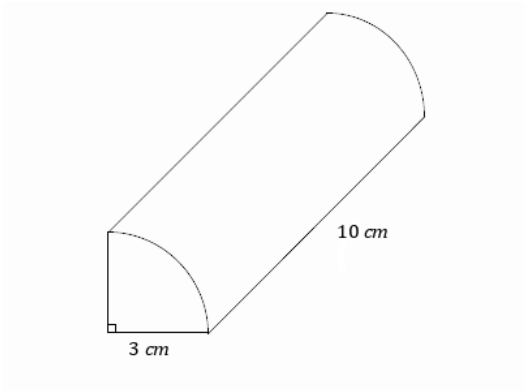
a



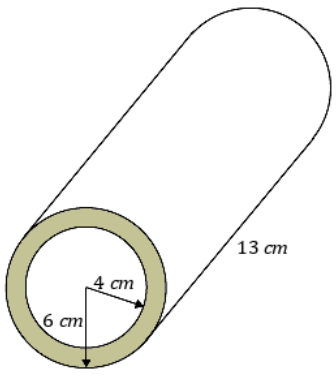
b



c

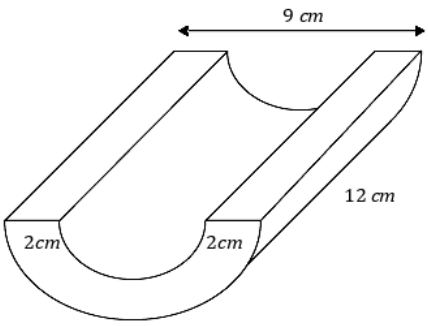


d

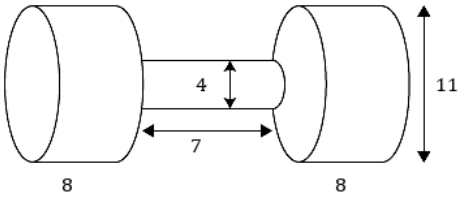


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

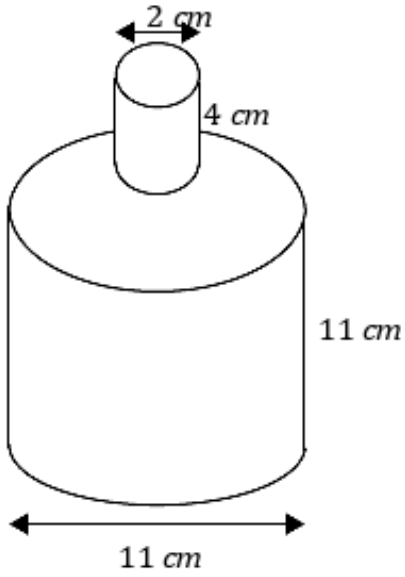
e



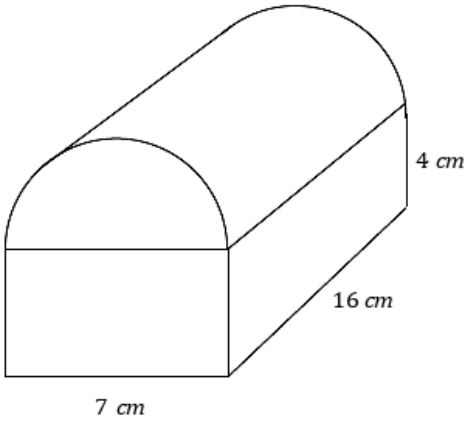
f



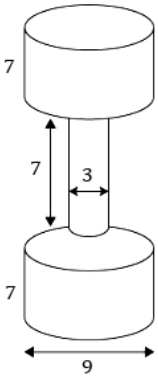
g



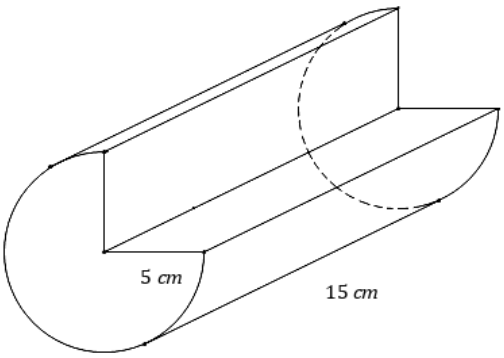
h



i

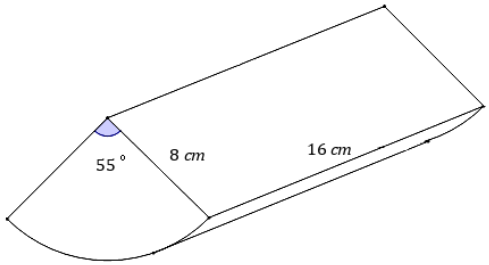


j



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

k

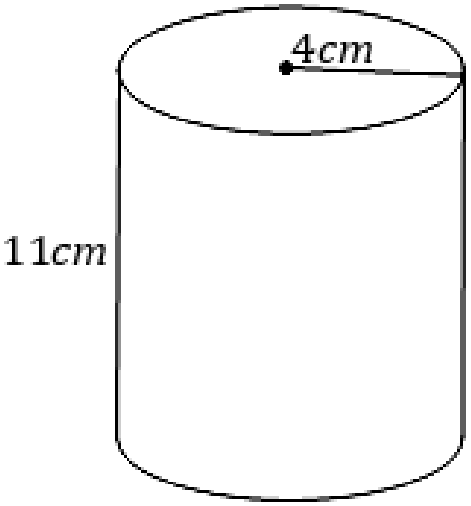


11 Xavier's mother told him to drink 4 glasses of water each day. She gave him a cylindrical glass with height 11 cm and radius 4 cm.

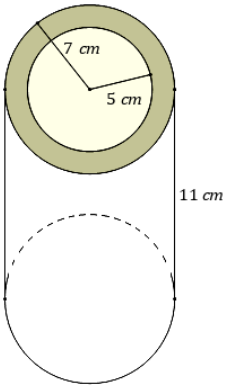
a Find the volume of the glass.
Round your answer to two decimal places.

b Assuming that he drinks 4 full glasses as his mother suggested, calculate the volume of water Xavier drinks each day.
Round your answer to two decimal places.

c If Xavier follows this drinking routine for a week, how many litres of water would he drink altogether?
Round your answer to one decimal place.

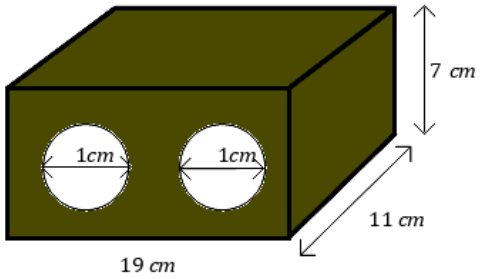


12 A cylinder is hollowed out by a hole of radius 5 cm. Find the volume of the remaining figure, correct to two decimal places.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13 We wish to calculate the volume of the solid shown by considering it as a prism. The cylinders are drilled perpendicular to the surface.



a Calculate the area of the front face correct to two decimal places.

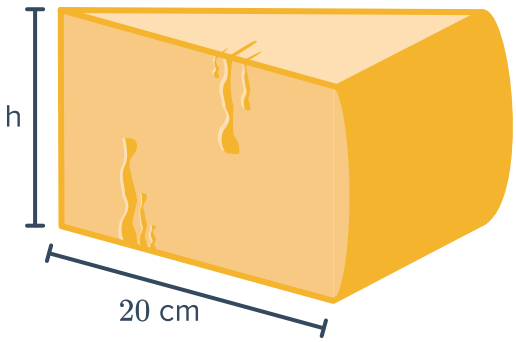
b Hence find the volume of the prism.

Additional questions

14 Queso Fresco is sold in wheels. The volume of a wheel of Queso Fresco is $2002\ \text{cm}^3$: The height of a wheel of Queso Fresco is $6\ \text{cm}$, find the radius.



15 Parmigiano-Reggiano is formed in wheels, and then cut into twenty equal slices before being packaged and sold. If the volume of a certain wheel of Parmigiano-Reggiano is $8000\pi\ \text{cm}^3$:



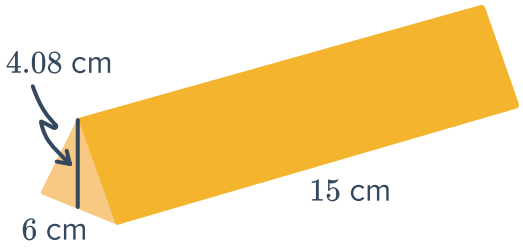
a Find the volume of a slice of Parmigiano-Reggiano that has been packaged for sale.

b Find the height of the wheel of Parmigiano-Reggiano.

16 A standard six-sided die has a side length of $5\ \text{mm}$. If Kyle keeps his die in a pouch with a volume of $1000\ \text{mm}^3$, how many die can he fit in the pouch?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

17 A chocolate company sells their bars in a special triangular prism packaging:



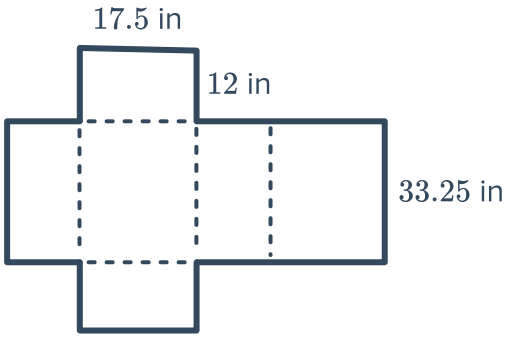
a Find the volume of the chocolate packaging.

b The chocolate only fills 85% of the packaging. Find the volume of one bar of chocolate.

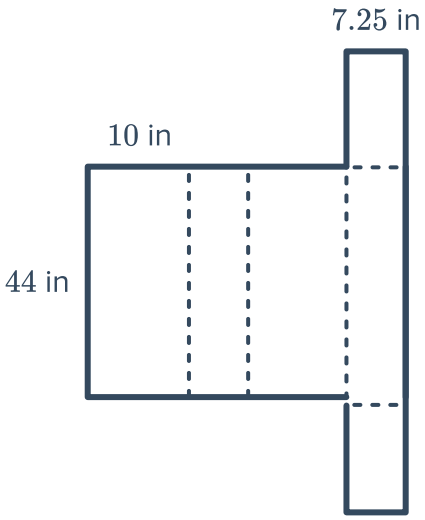
18 Tarpon Springs Aquarium is designing a new Reef Shark exhibit. In order to house the sharks they need a tank with a volume of $128\,000$ gallons.
In order to prevent the salt water from spilling over the edge of the tank, there needs to be a 1.5 meter clearance from the water level to the top of the shark tank. Using the conversion $1\ \text{m}^3 = 244\ \text{gal}$, design a rectangular tank that can hold $128\,000$ gallons of salt water.

19 U-Pack Moving Company sells two sizes of moving boxes. U-Pack claims to keep their prices lower than competitors by selling their boxes flat, and having customers assemble the boxes themselves.

Box 1 sells for $\$2.29$:



Box 2 sells for $\$0.99$:



If you have $60\ \text{ft}^3$ of stuff to pack into boxes for your move, which of the following combinations of boxes would you buy? Explain.

a $15\times$ Box 1

a $36\times$ Box 2

© ClipSAT: By Mr. Mohamed Abdallah +966597688647